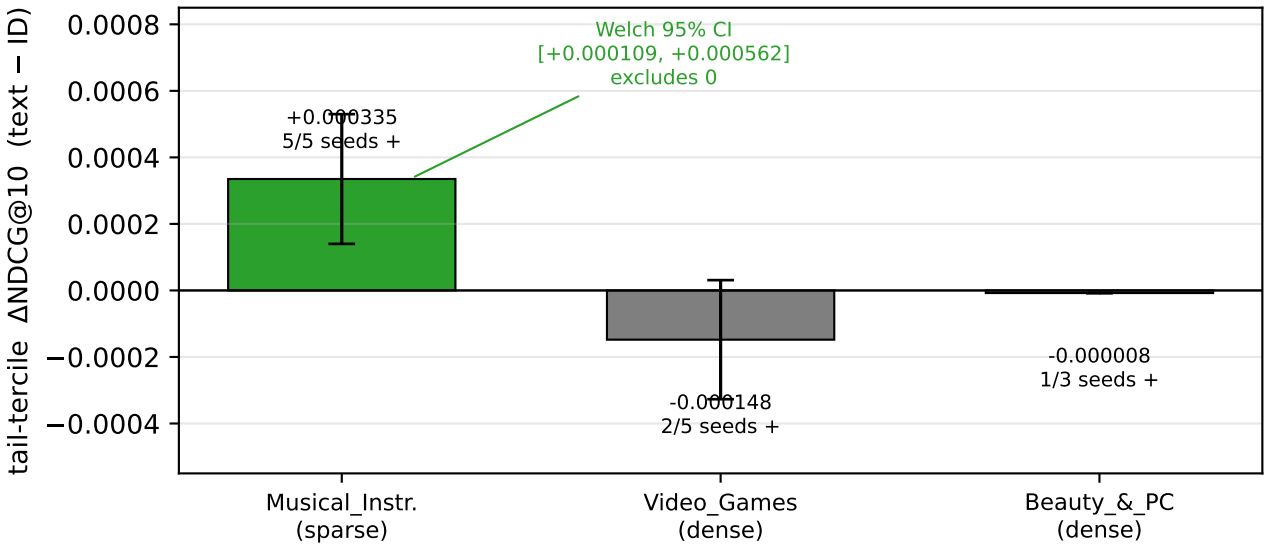
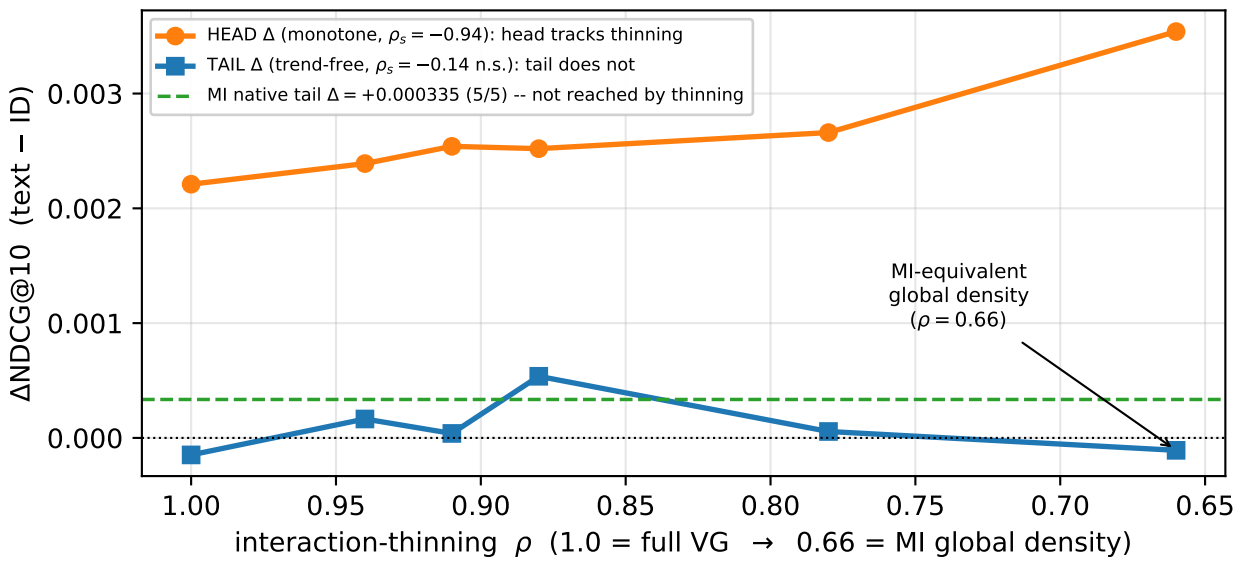


Dataset-conditional long-tail pattern and its two-axis descriptive contrast
(bundled interventions; one fixed draw; AR2023 5-core LLOO, full-catalog, NDCG@10;
all values 5-seed best-by-val except Beauty 2-3 seed)

(A) The dataset-conditional tail pattern
(sparse catalog: freq-5-heavy tail advantage)



(B) Interaction-density titration
(head tracks thinning / tail does not -- level contrasts)



(C) Two-axis descriptive contrast
(user-thinned point shifts positive (sugg.); count-thinned does not)

